$$Y_u{}^{i1\_i2\_} \to s_\gamma\, y_u{}^{i1i2} +$$

$$\frac{1}{16\,\pi^2}\,\Big(\frac{1}{144}\,s_\gamma\,\frac{1}{m_\Phi{}^2}\,\big(18\,y_u{}^{pi2}\,\big(\overline{y_d}^{pr}\,y_d{}^{i1r}\,\big(-2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,\big(1+s_\gamma{}^2\big)\big)\big)+3\,m_\Phi{}^2\,\overline{y_u}^{pr}\,y_u{}^{i1r}\,\big(1+c_\gamma{}^2\big)\big)+$$

$$y_u{}^{i1i2}\,\big(m_\Phi{}^2\,\big(-96\,g_3{}^2+27\,g_2{}^2\,\big(3+2\,c_\gamma{}^2+2\,s_\gamma{}^2\big)+g_1{}^2\,\big(19+18\,c_\gamma{}^2+18\,s_\gamma{}^2\big)\big)-$$

$$216\,C_{H^2}\,c_\gamma{}^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\big)\big)+\frac{1}{2}\sum_p s_{2\,\gamma}\,c_\gamma\,g_1{}^2\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(-C_{H^2}+m_\Phi{}^2\big)\,LF_{1,0}\big[m_{\tilde d}{}^p\big]+$$

$$\frac{3}{2}\,s_{2\,\gamma}\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,\overline{y_d}^{pr}\,y_d{}^{pr}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,LF_{1,0}\big[m_{\tilde d}{}^r\big]+\frac{1}{2}\sum_p s_{2\,\gamma}\,c_\gamma\,g_1{}^2\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}$$

$$\big(-C_{H^2}+m_\Phi{}^2\big)\,LF_{1,0}\big[m_{\tilde e}{}^p\big]+\frac{1}{2}\,s_{2\,\gamma}\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,\overline{y_e}^{pr}\,y_e{}^{pr}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,LF_{1,0}\big[m_{\tilde e}{}^r\big]+$$

$$\frac{1}{2}\,s_{2\,\gamma}\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,\big(\overline{y_e}^{pr}\,y_e{}^{pr}+\sum_p g_1{}^2\big)\,LF_{1,0}\big[m_{\tilde l}{}^p\big]+$$

$$\frac{1}{2}\,s_{2\,\gamma}\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(-C_{H^2}+m_\Phi{}^2\big)\,\big(-3\,\overline{y_d}^{pr}\,y_d{}^{pr}+3\,\overline{y_u}^{pr}\,y_u{}^{pr}+\sum_p g_1{}^2\big)\,LF_{1,0}\big[m_{\tilde q}{}^p\big]+$$

$$\sum_p s_{2\,\gamma}\,c_\gamma\,g_1{}^2\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,LF_{1,0}\big[m_{\tilde u}{}^p\big]+$$

$$\frac{3}{2}\,s_{2\,\gamma}\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,\overline{y_u}^{pr}\,y_u{}^{pr}\,y_u{}^{i1i2}\,\big(-C_{H^2}+m_\Phi{}^2\big)\,LF_{1,0}\big[m_{\tilde u}{}^r\big]+$$

$$\frac{3}{8}\,s_{4\,\gamma}\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(g_1{}^2+g_2{}^2\big)\,\big(-C_{H^2}+m_\Phi{}^2\big)\,LF_{1,0}[m_\Phi]-$$

$$\frac{1}{4}\,s_\gamma\,y_u{}^{pi2}\,\big(\overline{y_d}^{pr}\,y_d{}^{i1r}\,\big(4\,c_\gamma{}^2+s_\gamma{}^2\big)+3\,c_\gamma{}^2\,\overline{y_u}^{pr}\,y_u{}^{i1r}\big)\,LF_{1,1}[m_\Phi]-$$

$$\frac{1}{2}\,s_\gamma\,C_{H^2}\,c_\gamma{}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,y_u{}^{pi2}\,LF_{1,2}[m_\Phi]-\frac{1}{36}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{1,1,0}\big[m_1,\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{1}{72}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}\big[m_1,\,m_{\tilde q}{}^{i1}\big]-\frac{4}{9}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{1,1,0}\big[m_1,\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{2}{9}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}\big[m_1,\,m_{\tilde u}{}^{i2}\big]+m_1\,\tilde\mu\,c_\gamma\,g_1{}^2\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,\big(c_\gamma{}^2-s_\gamma{}^2\big)\,LF_{1,1,0}[m_1,\,\tilde\mu]-$$

$$\frac{3}{4}\,s_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{1,1,0}\big[m_2,\,m_{\tilde q}{}^{i1}\big]+\frac{3}{8}\,s_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}\big[m_2,\,m_{\tilde q}{}^{i1}\big]+$$

$$3\,m_2\,\tilde\mu\,c_\gamma\,g_2{}^2\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,\big(c_\gamma{}^2-s_\gamma{}^2\big)\,LF_{1,1,0}[m_2,\,\tilde\mu]-$$

$$\frac{4}{3}\,s_\gamma\,g_3{}^2\,y_u{}^{i1i2}\,LF_{1,1,0}\big[m_3,\,m_{\tilde q}{}^{i1}\big]+\frac{2}{3}\,s_\gamma\,g_3{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}\big[m_3,\,m_{\tilde q}{}^{i1}\big]-$$

$$\frac{4}{3}\,s_\gamma\,g_3{}^2\,y_u{}^{i1i2}\,LF_{1,1,0}\big[m_3,\,m_{\tilde u}{}^{i2}\big]+\frac{2}{3}\,s_\gamma\,g_3{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}\big[m_3,\,m_{\tilde u}{}^{i2}\big]+$$

$$3\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,\big(s_\gamma\,a_d{}^{pr}+\tilde\mu\,c_\gamma\,y_d{}^{pr}\big)\,LF_{1,1,0}\big[m_{\tilde d}{}^r,\,m_{\tilde q}{}^p\big]-$$

$$\frac{1}{2}\,s_\gamma\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,y_u{}^{pi2}\,LF_{1,1,0}\big[m_{\tilde d}{}^r,\,\tilde\mu\big]+$$

$$c_\gamma\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,\big(c_\gamma\,\overline{a_e}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_e}^{pr}\big)\,\big(s_\gamma\,a_e{}^{pr}+\tilde\mu\,c_\gamma\,y_e{}^{pr}\big)\,LF_{1,1,0}\big[m_{\tilde e}{}^r,\,m_{\tilde l}{}^p\big]-$$

$$\frac{1}{2}\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(c_\gamma\,\overline{a_e}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_e}^{pr}\big)\,\big(s_\gamma\,c_\gamma\,a_e{}^{pr}\,\big(2\,C_{H^2}-m_\Phi{}^2\big)+\tilde\mu\,y_e{}^{pr}\,\big(2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,s_\gamma{}^2\big)\big)$$

$$LF_{2,1,0}\big[m_{\tilde l}{}^p,\,m_{\tilde e}{}^r\big]+\frac{1}{2}\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(c_\gamma\,\overline{a_e}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_e}^{pr}\big)$$

$$\big(s_\gamma\,c_\gamma\,a_e{}^{pr}\,\big(2\,C_{H^2}-m_\Phi{}^2\big)+\tilde\mu\,y_e{}^{pr}\,\big(2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,s_\gamma{}^2\big)\big)\,LF_{3,1,-1}\big[m_{\tilde l}{}^p,\,m_{\tilde e}{}^r\big]+$$

$$s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(-c_\gamma\,\overline{a_e}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_e}^{pr}\big)\,\big(-c_\gamma\,a_e{}^{pr}+s_\gamma\,\tilde\mu\,y_e{}^{pr}\big)\,LF_{3,1,0}\big[m_{\tilde l}{}^p,\,m_{\tilde e}{}^r\big]-$$

$$3\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(-c_\gamma\,\overline{a_e}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_e}^{pr}\big)\,\big(-c_\gamma\,a_e{}^{pr}+s_\gamma\,\tilde\mu\,y_e{}^{pr}\big)\,LF_{4,1,-1}\big[m_{\tilde l}{}^p,\,m_{\tilde e}{}^r\big]+$$

$$2\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(-c_\gamma\,\overline{a_e}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_e}^{pr}\big)\,\big(-c_\gamma\,a_e{}^{pr}+s_\gamma\,\tilde\mu\,y_e{}^{pr}\big)\,LF_{5,1,-2}\big[m_{\tilde l}{}^p,\,m_{\tilde e}{}^r\big]-$$

$$\frac{3}{2}\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,\big(s_\gamma\,c_\gamma\,a_d{}^{pr}\,\big(2\,C_{H^2}-m_\Phi{}^2\big)+\tilde\mu\,y_d{}^{pr}\,\big(2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,s_\gamma{}^2\big)\big)$$

$$LF_{2,1,0}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r\big]+\frac{3}{2}\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)$$

$$\big(s_\gamma\,c_\gamma\,a_d{}^{pr}\,\big(2\,C_{H^2}-m_\Phi{}^2\big)+\tilde\mu\,y_d{}^{pr}\,\big(2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,s_\gamma{}^2\big)\big)\,LF_{3,1,-1}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r\big]+$$

$$3\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(-c_\gamma\,\overline{a_d}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,\big(-c_\gamma\,a_d{}^{pr}+s_\gamma\,\tilde\mu\,y_d{}^{pr}\big)\,LF_{3,1,0}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r\big]-$$

$$9\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(-c_\gamma\,\overline{a_d}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,\big(-c_\gamma\,a_d{}^{pr}+s_\gamma\,\tilde\mu\,y_d{}^{pr}\big)\,LF_{4,1,-1}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r\big]+$$

$$6\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(-c_\gamma\,\overline{a_d}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,\big(-c_\gamma\,a_d{}^{pr}+s_\gamma\,\tilde\mu\,y_d{}^{pr}\big)\,LF_{5,1,-2}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r\big]+$$

$$3\,c_\gamma\,\frac{1}{m_\Phi{}^4}\,y_u{}^{i1i2}\,\big(C_{H^2}-m_\Phi{}^2\big)\,\big(c_\gamma\,\overline{a_u}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_u}^{pr}\big)\,\big(-s_\gamma\,a_u{}^{pr}+\tilde\mu\,c_\gamma\,y_u{}^{pr}\big)\,LF_{1,1,0}\big[m_{\tilde q}{}^p,\,m_{\tilde u}{}^r\big]-$$

$$s_\gamma\,\overline{y_u}^{pr}\,y_u{}^{pi2}\,y_u{}^{i1r}\,LF_{1,1,0}\big[m_{\tilde q}{}^p,\,\tilde\mu\big]+\frac{3}{2}\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}$$

$$\big(\overline{a_u}^{pr}\,\big(2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,s_\gamma{}^2\big)+s_\gamma\,\tilde\mu\,c_\gamma\,\overline{y_u}^{pr}\,\big(2\,C_{H^2}-m_\Phi{}^2\big)\big)\,\big(s_\gamma\,a_u{}^{pr}-\tilde\mu\,c_\gamma\,y_u{}^{pr}\big)\,LF_{2,1,0}\big[m_{\tilde u}{}^r,\,m_{\tilde q}{}^p\big]+$$

$$\frac{3}{2}\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(\overline{a_u}^{pr}\,\big(2\,C_{H^2}\,c_\gamma{}^2+m_\Phi{}^2\,s_\gamma{}^2\big)+s_\gamma\,\tilde\mu\,c_\gamma\,\overline{y_u}^{pr}\,\big(2\,C_{H^2}-m_\Phi{}^2\big)\big)$$

$$\big(-s_\gamma\,a_u{}^{pr}+\tilde\mu\,c_\gamma\,y_u{}^{pr}\big)\,LF_{3,1,-1}\big[m_{\tilde u}{}^r,\,m_{\tilde q}{}^p\big]+$$

$$3\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(s_\gamma\,\overline{a_u}^{pr}-\tilde\mu\,c_\gamma\,\overline{y_u}^{pr}\big)\,\big(s_\gamma\,a_u{}^{pr}-\tilde\mu\,c_\gamma\,y_u{}^{pr}\big)\,LF_{3,1,0}\big[m_{\tilde u}{}^r,\,m_{\tilde q}{}^p\big]-$$

$$9\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(s_\gamma\,\overline{a_u}^{pr}-\tilde\mu\,c_\gamma\,\overline{y_u}^{pr}\big)\,\big(s_\gamma\,a_u{}^{pr}-\tilde\mu\,c_\gamma\,y_u{}^{pr}\big)\,LF_{4,1,-1}\big[m_{\tilde u}{}^r,\,m_{\tilde q}{}^p\big]+$$

$$6\,s_\gamma\,C_{H^2}\,y_u{}^{i1i2}\,\big(s_\gamma\,\overline{a_u}^{pr}-\tilde\mu\,c_\gamma\,\overline{y_u}^{pr}\big)\,\big(s_\gamma\,a_u{}^{pr}-\tilde\mu\,c_\gamma\,y_u{}^{pr}\big)\,LF_{5,1,-2}\big[m_{\tilde u}{}^r,\,m_{\tilde q}{}^p\big]-$$

$$\frac{1}{2}\,s_\gamma\,\overline{y_u}^{pr}\,y_u{}^{pi2}\,y_u{}^{i1r}\,LF_{1,1,0}\big[m_{\tilde u}{}^r,\,\tilde\mu\big]-\frac{3}{4}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}[\tilde\mu,\,m_1]-m_1\,\tilde\mu\,c_\gamma\,g_1{}^2\,\frac{1}{m_\Phi{}^2}$$

$$y_u{}^{i1i2}\,\big(C_{H^2}\,c_\gamma{}^2+s_\gamma{}^2\,\big(-C_{H^2}+m_\Phi{}^2\big)\big)\,LF_{2,1,0}[\tilde\mu,\,m_1]+\frac{1}{2}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{3,1,-2}[\tilde\mu,\,m_1]+$$

$$g_1{}^2\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(-2\,s_\gamma\,C_{H^2}\,m_\Phi{}^2+m_1\,\tilde\mu\,c_\gamma\,\big(C_{H^2}\,c_\gamma{}^2+s_\gamma{}^2\,\big(-C_{H^2}+m_\Phi{}^2\big)\big)\big)\,LF_{3,1,-1}[\tilde\mu,\,m_1]-$$

$$2\,m_1\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_1{}^2\,s_\gamma{}^2\,y_u{}^{i1i2}\,LF_{3,1,0}[\tilde\mu,\,m_1]+4\,s_\gamma\,C_{H^2}\,g_1{}^2\,y_u{}^{i1i2}\,LF_{4,1,-2}[\tilde\mu,\,m_1]+$$

$$6\,m_1\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_1{}^2\,s_\gamma{}^2\,y_u{}^{i1i2}\,LF_{4,1,-1}[\tilde\mu,\,m_1]-2\,s_\gamma\,C_{H^2}\,g_1{}^2\,y_u{}^{i1i2}\,LF_{5,1,-3}[\tilde\mu,\,m_1]-$$

$$4\,m_1\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_1{}^2\,s_\gamma{}^2\,y_u{}^{i1i2}\,LF_{5,1,-2}[\tilde\mu,\,m_1]-\frac{9}{4}\,s_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{2,1,-1}[\tilde\mu,\,m_2]-3\,m_2\,\tilde\mu\,c_\gamma\,g_2{}^2$$

$$\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(C_{H^2}\,c_\gamma{}^2+s_\gamma{}^2\,\big(-C_{H^2}+m_\Phi{}^2\big)\big)\,LF_{2,1,0}[\tilde\mu,\,m_2]+\frac{3}{2}\,s_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{3,1,-2}[\tilde\mu,\,m_2]+$$

$$3\,g_2{}^2\,\frac{1}{m_\Phi{}^2}\,y_u{}^{i1i2}\,\big(-2\,s_\gamma\,C_{H^2}\,m_\Phi{}^2+m_2\,\tilde\mu\,c_\gamma\,\big(C_{H^2}\,c_\gamma{}^2+s_\gamma{}^2\,\big(-C_{H^2}+m_\Phi{}^2\big)\big)\big)\,LF_{3,1,-1}[\tilde\mu,\,m_2]-$$

$$6\,m_2\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_2{}^2\,s_\gamma{}^2\,y_u{}^{i1i2}\,LF_{3,1,0}[\tilde\mu,\,m_2]+12\,s_\gamma\,C_{H^2}\,g_2{}^2\,y_u{}^{i1i2}\,LF_{4,1,-2}[\tilde\mu,\,m_2]+$$

$$18\,m_2\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_2{}^2\,s_\gamma{}^2\,y_u{}^{i1i2}\,LF_{4,1,-1}[\tilde\mu,\,m_2]-6\,s_\gamma\,C_{H^2}\,g_2{}^2\,y_u{}^{i1i2}\,LF_{5,1,-3}[\tilde\mu,\,m_2]-$$

$$12\,m_2\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_2{}^2\,s_\gamma{}^2\,y_u{}^{i1i2}\,LF_{5,1,-2}[\tilde\mu,\,m_2]+\frac{1}{4}\,s_\gamma\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,y_u{}^{pi2}\,LF_{2,1,-1}\big[\tilde\mu,\,m_{\tilde d}{}^r\big]+$$

$$\frac{1}{2}\,s_\gamma\,\overline{y_u}^{pr}\,y_u{}^{pi2}\,y_u{}^{i1r}\,LF_{2,1,-1}\big[\tilde\mu,\,m_{\tilde q}{}^p\big]+\frac{1}{4}\,s_\gamma\,\overline{y_u}^{pr}\,y_u{}^{pi2}\,y_u{}^{i1r}\,LF_{2,1,-1}\big[\tilde\mu,\,m_{\tilde u}{}^r\big]+$$

$$\frac{2}{9}\,m_1\,g_1{}^2\,\big(s_\gamma\,a_u{}^{i1i2}-\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{1,1,1,0}\big[m_1,\,m_{\tilde q}{}^{i1},\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{1}{18}\,m_1\,C_{H^2}\,g_1{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,2,1,-1}\big[m_1,\,m_{\tilde q}{}^{i1},\,m_{\tilde u}{}^{i2}\big]-$$

$$\frac{1}{6}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{1,1,1,-1}\big[m_1,\,m_{\tilde q}{}^{i1},\,\tilde\mu\big]-\frac{1}{6}\,m_1\,\tilde\mu\,c_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{1,1,1,0}\big[m_1,\,m_{\tilde q}{}^{i1},\,\tilde\mu\big]+$$

$$\frac{1}{18}\,m_1\,C_{H^2}\,g_1{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,2,1,-1}\big[m_1,\,m_{\tilde u}{}^{i2},\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{2}{3}\,s_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{1,1,1,-1}\big[m_1,\,m_{\tilde u}{}^{i2},\,\tilde\mu\big]+\frac{2}{3}\,m_1\,\tilde\mu\,c_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{1,1,1,0}\big[m_1,\,m_{\tilde u}{}^{i2},\,\tilde\mu\big]-$$

$$\frac{1}{12}\,s_\gamma\,C_{H^2}\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,2,1,-2}\big[m_1,\,\tilde\mu,\,m_{\tilde q}{}^{i1}\big]-\frac{1}{12}\,m_1\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,2,1,-1}\big[m_1,\,\tilde\mu,\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{1}{3}\,s_\gamma\,C_{H^2}\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,2,1,-2}\big[m_1,\,\tilde\mu,\,m_{\tilde u}{}^{i2}\big]+\frac{1}{3}\,m_1\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_1{}^2\,y_u{}^{i1i2}\,LF_{2,2,1,-1}\big[m_1,\,\tilde\mu,\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{3}{2}\,s_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{1,1,1,-1}\big[m_2,\,m_{\tilde q}{}^{i1},\,\tilde\mu\big]+\frac{3}{2}\,m_2\,\tilde\mu\,c_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{1,1,1,0}\big[m_2,\,m_{\tilde q}{}^{i1},\,\tilde\mu\big]+$$

$$\frac{3}{4}\,s_\gamma\,C_{H^2}\,g_2{}^2\,y_u{}^{i1i2}\,LF_{2,2,1,-2}\big[m_2,\,\tilde\mu,\,m_{\tilde q}{}^{i1}\big]+\frac{3}{4}\,m_2\,C_{H^2}\,\tilde\mu\,c_\gamma\,g_2{}^2\,y_u{}^{i1i2}\,LF_{2,2,1,-1}\big[m_2,\,\tilde\mu,\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{8}{3}\,m_3\,g_3{}^2\,\big(s_\gamma\,a_u{}^{i1i2}-\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{1,1,1,0}\big[m_3,\,m_{\tilde q}{}^{i1},\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{2}{3}\,m_3\,C_{H^2}\,g_3{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,2,1,-1}\big[m_3,\,m_{\tilde q}{}^{i1},\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{2}{3}\,m_3\,C_{H^2}\,g_3{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,2,1,-1}\big[m_3,\,m_{\tilde u}{}^{i2},\,m_{\tilde q}{}^{i1}\big]+$$

$$\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(-c_\gamma\,\overline{a_d}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{1,1,1,0}\big[m_{\tilde d}{}^r,\,m_{\tilde q}{}^p,\,\tilde\mu\big]+$$

$$\frac{1}{2}\,C_{H^2}\,\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(-c_\gamma\,\overline{a_d}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{2,1,1,0}\big[m_{\tilde d}{}^r,\,m_{\tilde q}{}^p,\,\tilde\mu\big]+$$

$$\frac{1}{2}\,C_{H^2}\,\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{3,1,1,-1}\big[m_{\tilde d}{}^r,\,m_{\tilde q}{}^p,\,\tilde\mu\big]+$$

$$\frac{1}{4}\,C_{H^2}\,\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{2,2,1,-1}\big[m_{\tilde d}{}^r,\,\tilde\mu,\,m_{\tilde q}{}^p\big]+$$

$$\frac{1}{2}\,C_{H^2}\,\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(-c_\gamma\,\overline{a_d}^{pr}+s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{2,1,1,0}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r,\,\tilde\mu\big]+$$

$$\frac{1}{2}\,C_{H^2}\,\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{3,1,1,-1}\big[m_{\tilde q}{}^p,\,m_{\tilde d}{}^r,\,\tilde\mu\big]+$$

$$\frac{1}{4}\,C_{H^2}\,\tilde\mu\,y_d{}^{i1r}\,y_u{}^{pi2}\,\big(c_\gamma\,\overline{a_d}^{pr}-s_\gamma\,\tilde\mu\,\overline{y_d}^{pr}\big)\,LF_{2,2,1,-1}\big[m_{\tilde q}{}^p,\,\tilde\mu,\,m_{\tilde d}{}^r\big]+$$

$$\frac{1}{9}\,m_1\,C_{H^2}\,g_1{}^2\,\big(s_\gamma\,a_u{}^{i1i2}-\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,1,1,0}\big[m_{\tilde q}{}^{i1},\,m_1,\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{1}{9}\,m_1\,C_{H^2}\,g_1{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{3,1,1,-1}\big[m_{\tilde q}{}^{i1},\,m_1,\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{4}{3}\,m_3\,C_{H^2}\,g_3{}^2\,\big(s_\gamma\,a_u{}^{i1i2}-\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,1,1,0}\big[m_{\tilde q}{}^{i1},\,m_3,\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{4}{3}\,m_3\,C_{H^2}\,g_3{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{3,1,1,-1}\big[m_{\tilde q}{}^{i1},\,m_3,\,m_{\tilde u}{}^{i2}\big]+$$

$$\frac{1}{9}\,m_1\,C_{H^2}\,g_1{}^2\,\big(s_\gamma\,a_u{}^{i1i2}-\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,1,1,0}\big[m_{\tilde u}{}^{i2},\,m_1,\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{1}{9}\,m_1\,C_{H^2}\,g_1{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{3,1,1,-1}\big[m_{\tilde u}{}^{i2},\,m_1,\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{4}{3}\,m_3\,C_{H^2}\,g_3{}^2\,\big(s_\gamma\,a_u{}^{i1i2}-\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{2,1,1,0}\big[m_{\tilde u}{}^{i2},\,m_3,\,m_{\tilde q}{}^{i1}\big]+$$

$$\frac{4}{3}\,m_3\,C_{H^2}\,g_3{}^2\,\big(-s_\gamma\,a_u{}^{i1i2}+\tilde\mu\,c_\gamma\,y_u{}^{i1i2}\big)\,LF_{3,1,1,-1}\big[m_{\tilde u}{}^{i2},\,m_3,\,m_{\tilde q}{}^{i1}\big]\Big)$$